3 The Rules of Inference

Given a natural number n and $n+1$ propositions $p_0, p_1, p_2, ..., p_n$, the syntax of a typical rule of inference generally follows the schema shown below.[1]

$$p_0, p_1, ..., p_{n-1} \vdash p_n$$

[1] We will adopt the convention that any axioms we specify in our logical system are implicitly assumed on the left-hand side of the $\vdash$ symbol, even if we don't write them.

$\vdash$ Crucially, the rules of inference invoke the use of a new meta-symbol $\vdash$ separating a (possibly empty) list of propositions on the left from a proposition on the right. This is meant to denote that, by *assuming* all of the propositions p_0, p_1, …, p_{n-1} on the left, we can *prove* the proposition p_n on the right follows by relying only on:

1. any axioms we have assumed,
2. the rules of inference,
3. any other theorems we have already proven.

This symbol $\vdash$ is often called a *"turnstile"* because of its shape, but we often read it as *"proves"* due to its role in denoting *syntactic entailment*,[2] also known as *provability*.

[2] Note that this is a distinct concept from *semantic entailment*, which is usually denoted by the $\vDash$ symbol and speaks about *satisfiability* in terms of *truth values*.

When we have two propositions p and q such that $p \vdash q$ and $q \vdash p$, we can denote $\dashv\vdash$ this in a more compact way by writing $p \dashv\vdash q$.

3.1 Rules for Zeroth-Order Expressions

To make these rules easier to read at a glance, we will simplify our notation for the presentation of these rules of inference by adopting the following convention within this section: whenever a rule of inference in this section uses the symbols p, q, or r, we will understand those symbols to refer to *arbitrary propositions*.

Implication Rules

Rule 1: Implication Introduction.
$(p \vdash q) \vdash p \to q$ ⌟

Rule 2: Implication Elimination.
$p, p \to q \vdash q$ ⌟

Equivalence Rules

Rule 3: Biconditional Introduction.
$p, q \vdash p \leftrightarrow q$ ⌟

Rule 4: Biconditional Elimination.
$p \leftrightarrow q \vdash (p \to q) \wedge (q \to p)$ ⌟

Conjunction Rules

Rule 5: Conjunction Introduction.
$p, q \vdash p \wedge q$ ⌟

Rule 6: Conjunction Elimination.
$p \wedge q \vdash p$
$p \wedge q \vdash q$ ⌟

Disjunction Rules

Rule 7: Disjunction Introduction.
$p \vdash p \vee q$
$q \vdash p \vee q$ ⌟

Rule 8: Disjunction Elimination.
$(p \vee q), (p \to r), (q \to r) \vdash r$ ⌟

Negation Rules

Rule 9: Negation Introduction.
$(p \to q), (p \to \neg q) \vdash \neg p$ ⌟

Rule 10: Negation Elimination.
$\neg\neg p \vdash p$ ⌟

Truth Rules

Rule 11: Truth Introduction.
$\vdash \top$ ⌟

Rule 12: Law of the Excluded Middle.
$\vdash \neg p \vee p$ ⌟

Falsity Rules

Rule 13: False Introduction.
$\neg p \wedge p \vdash \bot$ ⌟

Rule 14: False Elimination.
$\bot \vdash p$ ⌟

3.2 Theorems of the Propositional Calculus

Theorem 3.1.
For any propositions p and q, we have the following deduction.

$$p \to q \ \vdash \ (p \ \vdash \ q)$$ ⌟

Theorem 3.2: Conditional Disintegration.
For any propositions p and q, we have the following equivalence.

$$p \to q \ \dashv\vdash \ \neg p \vee q$$ ⌟

Theorem 3.3: Contrapositive Equivalence.
For any propositions p and q, we have the following equivalence.

$$p \to q \quad \dashv\vdash \quad \neg q \to \neg p$$ ⌟

Theorem 3.4: Exportation.
For any propositions p, q, and r, we have the following equivalence.

$$(p \wedge q) \to r \quad \dashv\vdash \quad p \to (q \to r)$$ ⌟

Theorem 3.5: Absorption.
For any propositions p and q, we have the following equivalence.

$$p \to q \quad \dashv\vdash \quad p \to (p \to q)$$ ⌟

Theorem 3.6: Commutativity of Conjunctions and Disjunctions.
For any propositions p and q, the following equivalences hold.

$$p \wedge q \quad \dashv\vdash \quad q \wedge p$$
$$p \vee q \quad \dashv\vdash \quad q \vee p$$ ⌟

Theorem 3.7: Associativity of Conjunctions and Disjunctions.
For any propositions p, q, and r, the following equivalences hold.

$$p \wedge (q \wedge r) \quad \dashv\vdash \quad (p \wedge q) \wedge r$$
$$p \vee (q \vee r) \quad \dashv\vdash \quad (p \vee q) \vee r$$ ⌟

Theorem 3.8: Distributivity of Conjunctions and Disjunctions.
For any propositions p, q, and r, the following equivalences hold.

$$p \wedge (q \vee r) \quad \dashv\vdash \quad (p \wedge q) \vee (p \wedge r)$$
$$p \vee (q \wedge r) \quad \dashv\vdash \quad (p \vee q) \wedge (p \vee r)$$ ⌟

Theorem 3.9: Destructive Dilemma.
For any propositions p, q, r, and s the following equivalence holds.

$$(p \to r), (q \to s), (\neg r \vee \neg s) \quad \vdash \quad (\neg p \vee \neg q)$$ ⌟

Theorem 3.10: De Morgan's Theorems.
For any propositions p and q, the following two equivalences hold.

$$\neg(p \wedge q) \quad \dashv\vdash \quad \neg p \vee \neg q$$
$$\neg(p \vee q) \quad \dashv\vdash \quad \neg p \wedge \neg q$$ ⌟

3.3 *Rules for First-Order Expressions*

In this section, as in the previous section, we will adopt the convention that φ stands for an arbitrary well-formed formula in order to make the rules easier to read.

Universal Rules

Rule 15: Universal Introduction.
$\varphi(t)$ for an arbitrary term $t \vdash \forall x(\varphi(x))$ ⌟

Rule 16: Universal Elimination.

$\forall x(\varphi(x)) \vdash \varphi(t)$ for any term t ⌟

Existential Rules

Rule 17: Existential Introduction.
$\varphi(t)$ for some term $t \vdash \exists x(\varphi(x))$ ⌟

Rule 18: Existential Elimination.
$\exists x(\varphi(x)) \vdash \varphi(t)$ for some new term t that has not yet appeared in this context ⌟

3.4 *Theorems of the Predicate Calculus*

Theorem 3.11: Quantifier Shift.
The following equivalences regarding repeated quantifiers hold for any *wff* φ.

$$\begin{aligned} \forall x \forall y \big(\varphi(x,y)\big) &\dashv\vdash \forall y \forall x \big(\varphi(x,y)\big) \\ \exists x \exists y \big(\varphi(x,y)\big) &\dashv\vdash \exists y \exists x \big(\varphi(x,y)\big) \end{aligned}$$

We also have the following for any existentially quantified universal *wff* φ.

$$\exists x \forall y \big(\varphi(x,y)\big) \vdash \forall y \exists x \big(\varphi(x,y)\big)$$

However, we *cannot* generally infer an existentially quantified universal from a universally quantified existential. The invalidity of this kind of reasoning is stated below for any *wff* φ.

$$\forall x \exists y \big(\varphi(x,y)\big) \nvdash \exists y \forall x \big(\varphi(x,y)\big)$$ ⌟

Theorem 3.12: Negation of Quantifiers.
The following equivalences hold for any *wff* φ.

$$\begin{aligned} \neg \forall x \big(\varphi(x)\big) &\dashv\vdash \exists x \big(\neg \varphi(x)\big) \\ \neg \exists x \big(\varphi(x)\big) &\dashv\vdash \forall x \big(\neg \varphi(x)\big) \end{aligned}$$ ⌟

Theorem 3.13: Distribution of Quantifiers.
Let φ be an arbitrary *wff,* and let p be an arbitrary proposition that does not involve the symbol *"*x*"*. The following equivalences then hold—be mindful of parentheses.

$$\begin{aligned} \forall x \big(\varphi(x) \wedge p\big) &\dashv\vdash \forall x \big(\varphi(x)\big) \wedge p \\ \exists x \big(\varphi(x) \wedge p\big) &\dashv\vdash \exists x \big(\varphi(x)\big) \wedge p \end{aligned}$$

We also have similar equivalences for disjunctive statements.

$$\begin{aligned} \forall x \big(\varphi(x) \vee p\big) &\dashv\vdash \forall x \big(\varphi(x)\big) \vee p \\ \exists x \big(\varphi(x) \vee p\big) &\dashv\vdash \exists x \big(\varphi(x)\big) \vee p \end{aligned}$$

Conditional statements involving a quantifier in the *consequent* behave similarly.

$$\begin{aligned} \forall x \big(p \to \varphi(x)\big) &\dashv\vdash p \to \forall x \big(\varphi(x)\big) \\ \exists x \big(p \to \varphi(x)\big) &\dashv\vdash p \to \exists x \big(\varphi(x)\big) \end{aligned}$$

Conditional statements with a quantified *antecedent,* however, are more nuanced. Additional parentheses have been added for clarity.

$$\begin{aligned} \forall x \big(\varphi(x) \to p\big) &\dashv\vdash \Big(\exists x \big(\varphi(x)\big)\Big) \to p \\ \exists x \big(\varphi(x) \to p\big) &\dashv\vdash \Big(\forall x \big(\varphi(x)\big)\Big) \to p \end{aligned}$$ ⌟